

PROJECT ABSTRACT

RISC-V Based SoC with PWM, Watchdog Timer and Seven-Segment Display Controllers

1. Problem Statement:

Modern embedded systems require a compact and programmable architecture capable of performing computation, controlling peripherals, monitoring system operation, and providing user-readable output. Implementing these functions using separate hardware modules can increase system complexity and make integration and verification more difficult. Therefore, there is a need for a System-on-Chip (SoC) that integrates a general-purpose processor with memory, communication interfaces, control peripherals, and dedicated hardware IPs within a common architecture.

This project proposes the design and verification of a RISC-V based System-on-Chip integrating a RISC-V processor with instruction memory, data memory, bus interconnect, UART, Timer, and GPIO. Three additional IPs, namely a PWM Generator, Watchdog Timer, and Seven-Segment Display Controller, are integrated into the SoC to provide control, monitoring, and display functionality. The processor configures and controls these peripherals through the system bus and executes an application to demonstrate their operation. The complete system is implemented at RTL level and verified through simulation and functional testbenches.

2. Proposed SoC Architecture:

The proposed SoC is organized around the RISC-V processor as the central processing unit. The processor is connected to instruction memory, data memory, and peripheral IP blocks through a common bus interconnect. UART provides communication with a host computer, while the Timer and GPIO provide basic timing and general-purpose control functions. The additional IPs—PWM Generator, Watchdog Timer, and Seven-Segment Display Controller—are integrated as memory-mapped peripherals and are controlled by the processor.

The system follows this data-flow:

- **Processing** — The RISC-V processor executes the application program and controls the different peripherals.
- **Memory Access** — Instruction Memory stores the program instructions, while Data Memory stores application data and intermediate results.
- **PWM Generation** — The PWM Generator produces a configurable pulse-width-modulated signal based on parameters programmed by the processor.
- **System Monitoring** — The Watchdog Timer monitors system operation and provides a mechanism for detecting abnormal or stalled processor activity.
- **Display** — The Seven-Segment Display Controller receives data from the processor and drives the seven-segment display to provide a simple visual indication of system status or numerical values.
- **Communication and Control** — UART, Timer, and GPIO provide communication, timing, and general-purpose control capabilities within the SoC.

3. Selected Additional IPs:

PWM Generator — The PWM Generator is a hardware IP used to generate a pulse-width-modulated output signal. The processor configures the required parameters through control registers, allowing the

duty cycle and other operating parameters to be controlled digitally. The IP can be used for applications such as digital control and actuator interfacing.

Watchdog Timer — The Watchdog Timer is a monitoring IP designed to supervise the operation of the system. It uses a configurable timer and can detect when the processor fails to respond within the expected time. The processor periodically services or resets the watchdog during normal operation, demonstrating a basic system-monitoring mechanism.

Seven-Segment Display Controller — The Seven-Segment Display Controller is used to display numerical or status information generated by the processor. It receives the required display data through memory-mapped control registers and generates the corresponding segment control signals. This provides a simple interface for observing system status and output during operation.

The complete SoC is implemented using **RTL design** and verified through simulation and testbenches. The project demonstrates the integration of a RISC-V processor with multiple peripheral IPs through a common bus architecture and provides a modular platform that can be extended with additional hardware accelerators or peripheral controllers in future designs.